Note:

It is worth noting that while analyzing the program, it was observed that only case 1 is active and functioning correctly. This implies that the program solely performs the data exploration task as intended, yielding the desired outputs.

Report: Analysis of the C Program for Data Exploration

This report provides a detailed analysis of the C program designed for data exploration. The program reads data from a specific text file and determines the maximum and minimum limits of X and Y coordinates. Additionally, it calculates the minimum and maximum limits of data points belonging to different classes.

Program Structure:

The program contains a `main()` function. This function prompts the user for a choice within a loop and performs actions based on the selection using a `switch-case` structure. File operations are conducted using `fopen()` and `fclose()` functions. Data points are read using `fscanf()`.

Functionality of the Program:

1. Data Reading:

The program opens a text file named "data.txt" and reads data points. Each line contains X and Y coordinates and a class label. For each data point read, it calculates the minimum and maximum values of X and Y coordinates based on the respective class.

2. Calculation of Minimum and Maximum Values:

While iterating through the data points, the program updates the minimum and maximum values of X and Y coordinates for each class. Thus, it computes the minimum and maximum X and Y values for each class.

3. Printing Results:

The program prints the calculated minimum and maximum X and Y values to the screen. This is done separately for each class and presented to the user.

Performance and Improvement:

While reading data points, the program performs operations based on the class of each data point, which may be slow for large datasets. For a more efficient approach, data points could be separated into classes first, and then minimum and maximum values could be calculated for each class.

Additionally, the program's user interface is simple and user-friendly. However, allowing the user to input the file name could make it more flexible.

Conclusion:

This program serves as a basic tool for data exploration. It provides essential statistics (minimum and maximum X and Y coordinates) from the dataset to the user. However, there are potential improvements that could make it more efficient, especially for larger datasets.

OUTPUTS:

```
Menu:
1) Explore Data
2) Train Model
3) Test Model
4) Exit
Enter your choice:
1
Minimum and Maximum X and Y values:
Class 0: X in [1 48], Y in [0 45]
Class 1: X in [5 49], Y in [0 49]
```

```
Menu:
1) Explore Data
2) Train Model
3) Test Model
4) Exit
Enter your choice:
4
Exiting program...
```